Bridging the Theoretical and Experimental Biophysicists in Bangladesh
Networking Event Mini Symposium-2025
University of Barishal
17 April, 2025

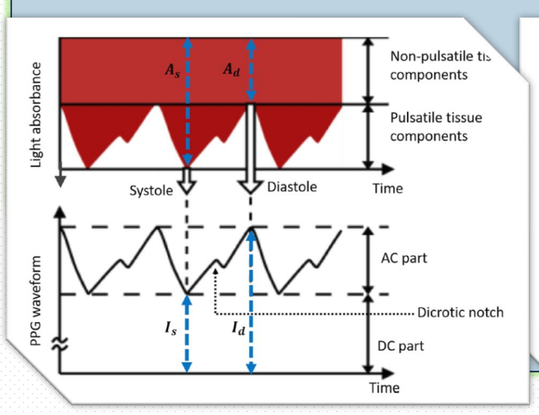
PPG-Based Assessment of PEMF Influence on Arterial Hemodynamics

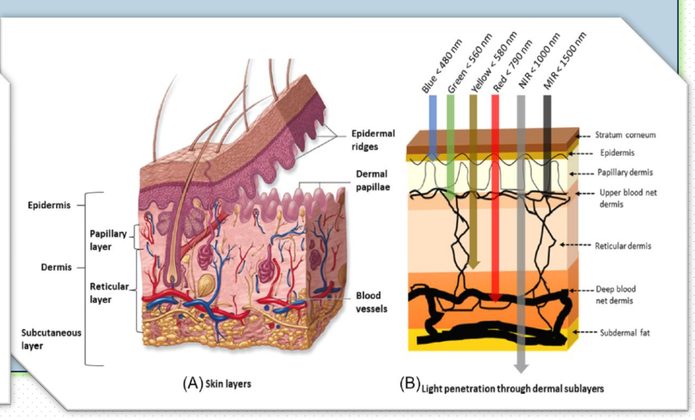
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Objectives

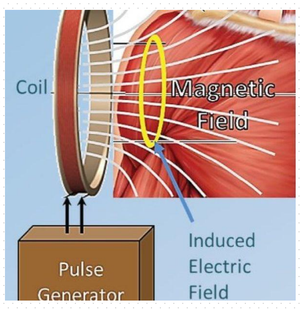
This study aimed to investigate the effects of various Pulsed Electromagnetic Field (PEMF) configurations on Arterial blood flow as measured by Photoplethysmography (PPG).

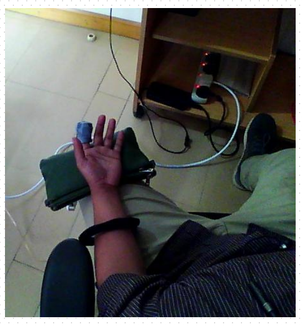
- ✓ To assess the effect of **PEMF** on peripheral arterial blood flow.
- ✓ To compare the impact of **different coil orientations (**parallel, perpendicular).
- ✓ To analyze changes in **PPG signals** under baseline and PEMF conditions.
- ✓ To evaluate **time-domain** and **frequency-domain** parameters.
- ✓ To identify coil positions that **enhance or reduce vascular resistance and flow**.







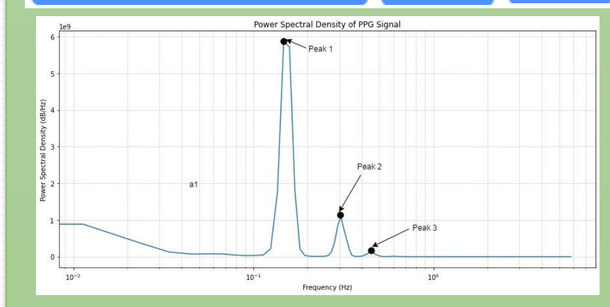


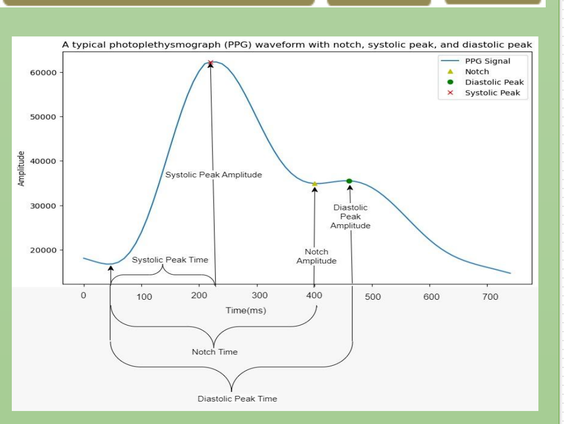


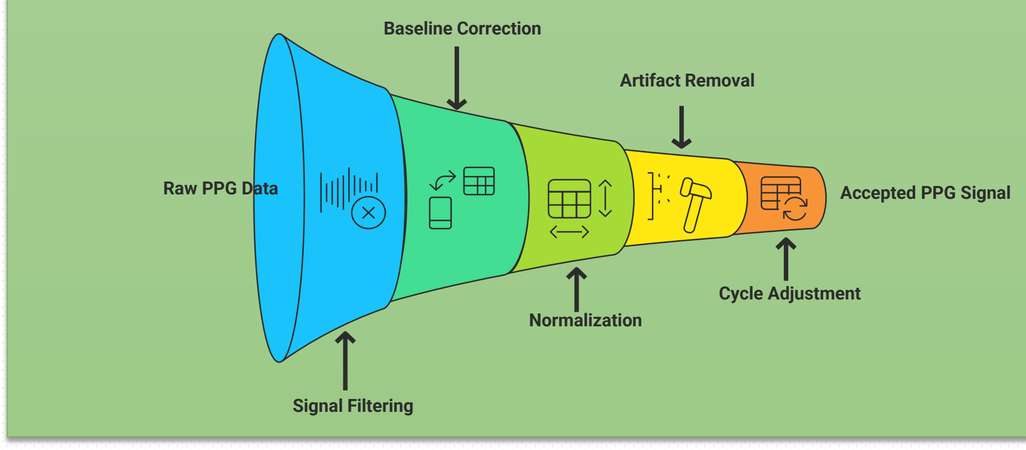
Result

Characteristic	Perpendicular	Parallel
Systolic Peak	↑	↓
Diastolic Peak	↓	↑
Systolic Area	↓	↑
Diastolic Area	↓	↑
Power Spectrum Peak-1	↓	↓
Power Spectrum Peak-2	↑	↑
Power Spectrum Peak-3	↑	↓

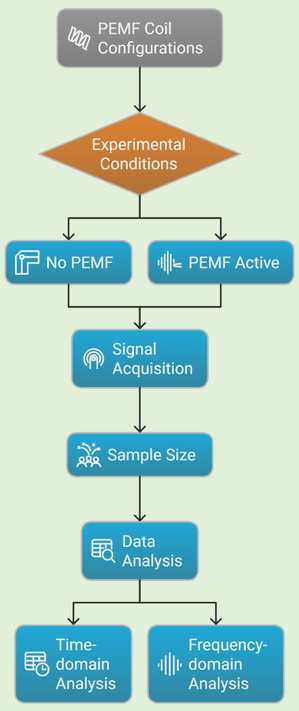
Characteristic	Forward	Reverse
Systolic Peak	↓	↑
Total Area	↓	↑
Diastolic Area	↓	↑
Power Spectrum Peak-2	↑	↑
Power Spectrum Peak-3	↑	↑
5 PS_Area (freq 2 to 5)	↓	↓







Methodology



PEMF Application:

- A PEMF coil was used in three configurations:

1. Oval shape over the brachial artery.
2. Parallel to the forearm (anterior below the elbow) over the radial artery in both clockwise and counterclockwise directions.
3. Perpendicular to the forearm (anterior below the elbow) covering the radial artery.

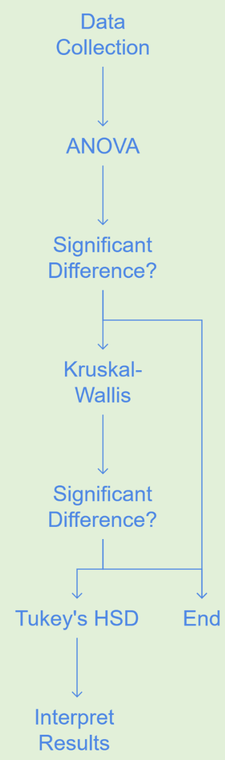
- Experimental Conditions:

Data were recorded under two conditions:

1. No PEMF (baseline)
2. PEMF active (10Hz-100µS-3.5A)

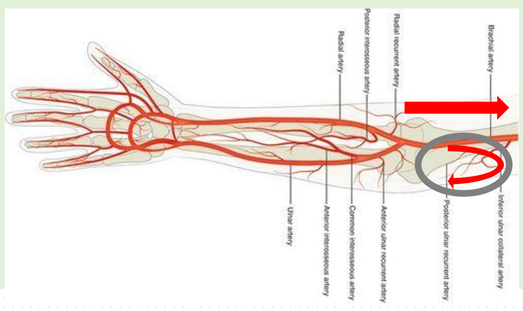
- Signal Acquisition: Photoplethysmogram (PPG) signals were captured from the index fingertips to monitor changes in peripheral blood flow.
- Sample Size: Data were collected from 43 arteries with proper consent.
- Data Analysis:

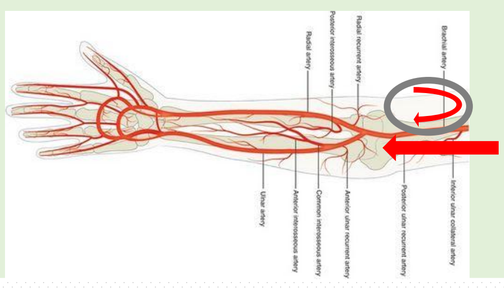
1. Time-domain parameters: systolic/diastolic peaks, areas, pulse interval, dicrotic notch, and total area.
2. Frequency-domain analysis: power spectrum peaks and band-specific area (0–2 Hz, 2–5 Hz).











Conclusion

This study demonstrates that Pulsed Electromagnetic Field (PEMF) therapy significantly influences peripheral arterial blood flow, with its effectiveness varying by coil orientation. The forward oval configuration showed potential for reducing vascular resistance, while the reverse and parallel orientations enhanced blood flow and perfusion. These findings suggest that PEMF, when applied in specific directions, can be a promising non-invasive method for improving micro-circulation. The results also show that the way the magnetic field is applied plays a big role in how blood vessels respond. The forward oval setup may help blood flow more easily by lowering resistance, while the reverse and clockwise configurations increased the time between pulses and boosted blood flow during relaxation. On the other hand, the anticlockwise setup had the opposite effect. When the field was applied parallel to the blood vessels, it encouraged vessel widening and better circulation. But when the field was perpendicular, it reduced the strength and amount of blood flow. These findings highlight how different PEMF directions can lead to different effects on blood flow and vessel function.

Acknowledgement:
This work is funded by North South university Research Fund